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Côte d'Ivoire cocoa cooperatives: data and methods v1.1

Data access

The full dataset can be visualised and downloaded here:

<https://trase.earth/explore/facilities-data/map?facilityTypeId=cote-d-ivoire-cocoa-cooperatives>.

How to cite the data

Guye, V., & French, N. (2026). Côte d'Ivoire cocoa cooperatives: data and methods (Version 1.1) [Data set]. Trase. <https://doi.org/10.48650/8SP9-FN57>

Dataset description

This dataset contains the cocoa cooperatives in Côte d'Ivoire and their attributes, including geographic coordinates, departments, names of the companies disclosing to buy from them in a given year, and supplying farmer base sizes. This database builds on the Cocoa Accountability Map, initially provided by Mighty Earth, and on further collected public disclosures by companies trading in cocoa or manufacturing cocoa products, and by certification bodies (only Rainforest Alliance currently) since 2020.

The data is provided as a unified spatial dataset available in different formats (e.g. GeoJSON, CSV, Excel), containing one record per uniquely identified cooperative per year.

Versioning

Increments in the first decimal of the version number (e.g., v1.0 to v1.1) reflect a database with the same variables built on additional supply chain information. An increment in the version integer reflects a structural change in the variables or datasets offered.

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Table 1: Version history

Version	Change & access
v1.1	Dataset updated to include information from the year 2024. Some additional location information was added across all years, and minor fixes were made to remove some duplicated data. https://doi.org/10.48650/8SP9-FN57
v1.0	Guye, Valentin, 2024, "Ivorian Cocoa Cooperatives and their Buyers (IC2B) – database v1.0." For the full source dataset and methodology, access: https://doi.org/10.14428/DVN/YBWJOR .

Table 2: Column descriptions

Column key	Name	Description	Data type
year	Year	The year of the data	year
abbreviated_name	Abbreviated cooperative name	The abbreviated name of the cooperative. When this is unknown, we repeat the full name.	string
full_name	Full cooperative name	The complete name of the cooperative. When this is unknown, we repeat the abbreviated name.	string
department_name	Department	The name of the department (Côte d'Ivoire administrative region) where the cooperative is located.	string

buyers	Buyers	List of commercial entities involved in the cocoa supply chain, in one or several of the trading, manufacturing or retailing levels. Companies may or may not self-disclose their supply chains.	string
buyers_list	Buyers (in JSON array format)	Same as buyers column but in JSON array format	JSON array
buyers_estimated_farmer_base	Buyers farmer base	Number of farmers estimated to supply the cooperative, as disclosed by the cooperative buyers.	number
coop_stable_id	Cooperative Stable ID	The unique integer identifier assigned to track a specific cooperative consistently across years.	number
department_id	Department Code	The standard administrative geocode for the department where the cooperative is located.	string
department_trase_id	Trase Department ID	The Trase-formatted administrative geocode for the department.	string
repeated_from_past_year	Repeated from Past Year	A boolean flag indicating if the flow or cooperative was rolled forward from a previous year's disclosure due to absence in current-year reporting.	boolean
geometry	Geometry	The spatial point data representing the cooperative's geographic location.	point

Definitions

Throughout this document, the following definitions apply:

- Cocoa cooperative/coop/co-operative refers to an entity disclosed to collect and sell cocoa in Côte d'Ivoire. In this context, no other type of supply chain actors are currently disclosed publicly. Therefore, no specific condition is imposed systematically on disclosed entities to qualify as a cooperative.
- Abbreviated name/abbreviated: a common attribute of Ivorian cooperatives (e.g. ACAD SCOOPS).
- Full name/full: the name of the cooperative in full form (e.g. ALLIANCE COOPERATIVE AGRICOLE DE DOUKOUYO).
- Geographic coordinates: set of geodesic longitude and latitude.
- Company/buyer: a commercial entity involved in the cocoa supply chain, in one or several of the trading, manufacturing or retailing levels. Companies may self-disclose their supply chains, or not.
- Link: A disclosed supply chain connection between a cooperative and a company sourcing from this cooperative in a given year. While individual links are not published as standalone rows in this dataset, they are used internally to calculate the cooperative's total estimated farmer base. Two links may represent the same flow of cocoa supply from one cooperative to several companies at vertically different stages in the supply chain.

Data collection

The database currently comprises data originating from public disclosures by the following companies (spelled this way in the data): ALBERT HEIJN, ALFRED RITTER, ALTER ECO, BARRY CALLEBAUT, BLOMMER, CARGILL, CEMOI, COLRUYT, ECOM, ECOOKIM, ETC GROUP, ETHIQUABLE, FERRERO, FILDISI, HERSHEY, MARS, MONDELEZ, NESTLE, OLAM, PURATOS, SUCDEN, TONY'S CHOCOLONELY, TOUTON, VALRHONA. Publicly available data from RAINFOREST ALLIANCE is also included. Note that publicly available data from Fairtrade is excluded as it is not relevant to this database.

Raw data is collected in three main ways: automatically pulling from online open access repositories, manually extracting PDF tables into spreadsheets, and manually extracting relevant information from reports available online. In addition, a substantial amount of data comes from the Cocoa Accountability Map (v3) (Mighty Earth, 2021) provided by Mighty Earth, recorded as valid for the baseline year 2019.

During data processing, commercial entities are categorized into different vertical stages. Disclosures are scrubbed to ensure product manufacturers (e.g., Mars, Nestlé, Ferrero, Tony's Chocolonely) are strictly separated from trading companies to prevent manufacturers from being incorrectly flagged as direct physical buyers at the cooperative level.

Cooperative identification & Stable IDs

The goal of this step is to take messy data from many different formats of company reports and turn it into a single, clean list of unique cooperatives. This is challenging because different cooperatives sometimes share the exact same name, or a single cooperative might be reported with slightly different GPS coordinates or slightly different name spelling/abbreviation. To handle this confusion, we clean the data in two phases: first a historical foundation was built, and from this we can update the data each year by matching newly-disclosed cooperatives to standardised naming and location assignment in the historical baseline.

1. The Historical Baseline (2019–2023)

To build the foundational list of unique cooperatives, we used abbreviated names, full names, and geographic coordinates. We consider that they distinguish cooperatives with decreasing levels of confidence, as follows: full names > coordinates > abbreviated names. Unique cooperatives were identified through a four-stage process:

- Pre-processing: Identifying attributes were standardized and coordinates were rounded to 3 decimals.
- Imputation: Missing information was iteratively imputed within groups of rows showing no sign of being distinct cooperatives.
- Homogenization: A common name was chosen for rows sharing both the same alternative name and identical coordinates.
- Differentiation: Highly rounded coordinates were used to distinguish homonym cooperatives only if they were located more than 11.1km apart.

2. Annual Updates (2024 onwards)

For new annual disclosures, records are mapped to the established historical Stable IDs using a hierarchical matching pipeline. The system attempts to match newly disclosed entities against the historical baseline using the following priority:

1. Full Name + Department District
2. Abbreviated Name + Department District
3. Full Name alone
4. Abbreviated Name alone

If a match is confirmed, the historical Cooperative Stable ID is assigned. If no match is found, the pipeline generates a brand new, original ID for the cooperative.

Geographic Sync & Imputation

A two-way geographic synchronization is applied to heal missing location data. Historical cooperatives lacking department classifications are updated using recent geographic disclosures. Conversely, newly disclosed cooperatives with missing spatial data inherit their coordinates and districts from their historical baseline match. Additionally, for the generation of the final spatial dataset (GeoJSON), any missing textual or categorical attributes are imputed using the nearest historical year where that specific cooperative had valid data.

We attribute a department name and identification number to cooperatives and links, based on a spatial join between cooperative coordinates and a shapefile of Ivorian departments. Where the department information is still not available, we infer it from a disclosed locality name. If company disclosures lack geographic information entirely, we conduct manual online searches for the specific cooperatives to pinpoint their department. We also extract town or city names embedded directly within the cooperative names themselves and match those locations to their corresponding administrative departments in Côte d'Ivoire.

The process described above is meant to improve the identification of cooperatives (including for past years) as more precise disclosures are issued by companies. This may lead to single identified cooperatives turning out to be several distinct cooperatives in future versions of this database. Although the opposite is less likely, it is not impossible that cooperatives currently identified as distinct may prove to be the same one. Moreover, brand new cooperatives can appear for the first time in future disclosures. Despite this instability in identifying attributes, stable coop IDs are granted across versions of the database. When a single coop turns out to be several distinct ones, the single's ID is deactivated from all subsequent versions of the database. The newly identified distinct coops get new, original IDs (as if they were disclosed for the first time). The same rationale applies if distinct coops turn out to be a single one.

Supply base estimate

In the dataset, we record the names of all known buyers from a cooperative in a given year, separated by commas, in the variable **buyers**. Companies recorded in this variable can be buyers at different vertical stages of the same cocoa flow.

Cooperative minimum size estimation

We impute the total number of farmers supplying a cooperative by summing up the farmer base sizes of its underlying disclosed links. To prevent double-counting supply bases that reflect the same physical flow of cocoa (e.g., disclosed by both a trader and a downstream manufacturer), we sum the underlying links up within each of the two kinds of buyers (traders vs. non-traders) and keep the maximum between both. Finally, we retain the maximum between this calculated number and the number of farmers certified under Rainforest Alliance. This final estimate is stored in the variable **buyers_estimated_farmer_base**.

We quantify the minimum overall size of the cooperatives in terms of the number of farmers. As in Renier et al. (2023), we assume the disclosed number of farmers refers strictly to those supplying that specific company (the link's farmer base), rather than the cooperative's total membership.

Historical Baseline Imputation (2019–2023)

Historically, missing underlying link sizes were imputed progressively: first from different sources for the same link in the same year, then from sizes of other links associated with the same cooperative. For links with still no size information, the baseline number was simulated for each cooperative using 1000 Monte Carlo estimates drawn from the set of disclosed cooperative sizes.

Annual Updates (2024 onwards)

If a cooperative's farmer count is missing from the current year's disclosures, the processing pipeline automatically imputes the missing value by rolling forward the maximum recorded farmer count associated with that specific cooperative's Stable ID from the historical baseline.

Inter-annual extrapolation

During annual updates, the pipeline identifies historical supply chain flows (based on the unique combination of Cooperative ID and Trader Name) that existed in the previous year but are missing in the current year's disclosures. These specific missing flows are rolled forward into the current year. Any extrapolated record in the dataset is flagged with the boolean dummy variable **repeated_from_past_year**.

References

Guye, V., (2024). Ivorian Cocoa Cooperatives and their Buyers (IC2B) – database v1.0, <https://doi.org/10.14428/DVN/YBWJOR>, Open Data @ UCLouvain

Renier, C. *et al.* (2023) ‘Transparency, traceability and deforestation in the Ivorian cocoa supply chain’, *Environmental Research Letters*, 18(2), p. 024030. Available at: <https://doi.org/10.1088/1748-9326/acad8e>.

Mighty Earth (2021) ‘Cocoa Accountability Map (Version 3.0)’. Available at: <https://mightyearth2022.users.earthengine.app/view/ghana-cocoa-accountability-map>

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